CHAPTER 5

Conditional probabilities and independence

Chapter overview

Objectives. Change reference universe; build a tree; use partition;
distinguish incompatibility and independence; treat up to four Bernoulli.
Prerequisites. Fragments, proportions, probabilities on a finite universe, intersection and meeting.
Annual benchmark. November 17·26: 8:00 a.m.

Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required.
diagnostic check is not graded.

1. Out of 40 students, 12 practise a given sport. Proportion?

2. If $P(A) = 0{,}3$, give $P(\overline{A})$.

3. Express "A and B" in like notation.

4. Express "A or B" with or inclusive.

5. Calculate $0{,}4 \times 0{,}25$.

6. A balanced dice is launched. $P(\{2; 4; 6\})$.

After the diagnostic check. Retrieve the concepts indicated with the teacher; repeat a question
and then explain the rule used.

Speaking and reasoning. Explain aloud what "knowing", "and" and "at least" mean.
a tree, justify addition and multiplication.

Search before formalizing

Build a table of 100 people for which A and B are independent, with 40 A and 30 B.
Change two boxes by keeping these margins to destroy independence.

Organization. Individual research 5 min, confrontation 2 5 min, pooling 5 min.
Keep a trace of a test, even if not completed. A conjecture must then be justified.

5.1 Condition: change denominator

Definition and results

For $P(A) > 0$, the probability of B knowing A is $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. In a table It is the effective quotient of $A \cap B$ A. In general $P_A(B) \neq P_B(A)$.

Status and justification. Definition. $P(A)$, we get the rule $P(A \cap B) = P(A)P_A(B)$ The condition $P(A) > 0$ is indispensable.

Example solved. Out of 100 people, 40 belong to A, 30 to B and 12 to both. $P_A(B) = \dfrac{12}{40} =$ 0,3, while $P_B(A) = \dfrac{12}{30} = 0{,}4$. $P(A \cap B) = 0{,}12$.

Point of vigilance

"Among the A" sets the denominator. A conditional probability is not an intersection.

Immediate application. Out of 60 objects, 20 are red; among the red, 5 are damaged. $P(R \cap D)$ and $P_R(D)$.

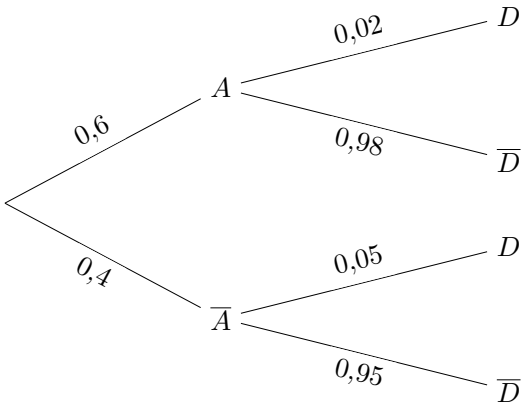
5.2 Trees, partitions and total probabilities

Definition and results

A partition $A_1, \ldots, A_k$ is a family of disjointed events whose meeting is the universe. If their probabilities are positive, $P(B) = \sum_i P(A_i)P_{A_i}(B)$ On a tree, probabilities from the same node have a sum of 1; the probabilities of a path multiply; the Disjointed paths meet by adding together.

Status and justification. B is the separate meeting of the $B \cap A_i$. By additivity and by definition of conditional probabilities, $P(B) = \sum_i P(B \cap A_i) = \sum_i P(A_i)P_{A_i}(B)$.

Example solved. Two machines provide 60% and 40% of production; their default rates 2 % and 5 %. $P(D) = 0{,}6 \times 0{,}02 + 0{,}4 \times 0{,}05 = 0{,}032$. Among the defects, the first machine Representative $\dfrac{0{,}012}{0{,}032} = \dfrac{3}{8}$.



A is the machine that provides 60 percent of the production and D a defect. For example, $0{,}02 = P_A(D)$: the probabilities of the second level are conditional.

Point of vigilance

The simple average of the rates is correct only if the weight of the branches is equal.

Immediate application. $P(A) = \frac{1}{3}$, $P_A(B) = \frac{3}{4}$, $P_{\overline{A}}(B) = \frac{1}{4}$. Calculate $P(B)$.

5.3 Independence and Incompatibility

Definition and results

A and B are independent if $P(A \cap B) = P(A)P(B)$ If $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$ They are incompatible if $A \cap B = \varnothing$. Two events of probability strictly positive and incompatible are not independent.

Status and justification. The conditional criterion results from a division by $P(A) > 0$. For events incompatible with positive probabilities, the left member is 0 and the product of right is strictly positive.

Example solved. On a balanced dice, A: even result, B: result at least 5. $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{6}$: A and "unfair result" are incompatible but not independent.

Point of vigilance

An independence in a model must be explicitly justified or laid down; "two events different" is not enough.

Immediate application. We give $P(A) = 0{,}4$, $P(B) = 0{,}5$, $P(A \cap B) = 0{,}18$. Are they independent?

5.4 Successive Tests and Bernoulli

Definition and results

A Bernoulli test has two results: success, probability p, and failure, probability 1 $-p$. In a repetition of independent and identical tests, with $n \leq 4$ here, probabilities branches remain p and 1 $-\ p$. A path of success is likely $p^k(1-p)^{n-k}$; on counts the paths that realize the event, without general theory of binomial distribution.

Status and justification. The probability of a path is a product, by independence. corresponding to an event are disjointed: their probabilities add up.

Example solved. For three tests with $p = \frac{1}{2}$, exactly two successes correspond to SSE, SES, ESS. Each path has a probability $\frac{1}{8}$, therefore total $\frac{3}{8}$. At least one success is likely $1 - (\frac{1}{2})^3 = \frac{7}{8}$.

Point of vigilance

Without a return to a ballot box, the probability generally changes after the first draw: Independent model can be false.

Immediate application. Four independent probability tests $p = \frac{1}{3}$. Calculate probability four failures and then at least one success.

5.5 Simulate and discuss the model

Definition and results

A pseudo-random simulation produces frequencies that fluctuate around a probability for a uniform point in the square [0; 1] ^{2}, the proportion checking $y \leq x^2$ estimates under the parabola; the proportion checking $x^2 + y^2 \leq 1$, multiplied by 4, estimates π. The repeats must use new prints.

Status and justification. Accepted and experienced simulation principle. is an interpretation of the uniform drawing; integration is not a prerequisite. Doesn't show an exact value.

Example solved. Among 10,000 points, if 7840 are in the quarter disk, the estimate of π value $\frac{4 \times 7840}{10000} = 3{,}136$. Two tests of the same size usually give different values.

Point of vigilance

Fixing a seed makes it possible to reproduce a test, not to remove the alea from the model or from the ensure accuracy.

Immediate application. A simulation of 2000 points gives 675 points under $y = x^2$. What Does she think so?

Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 5.1 · A table

A survey of 100 users of two transport services A and B. Of these, 40 use
A, 35 use B and 15 use both.
the same probability of being chosen. A and B meaning respectively "user uses the
service A" and "user uses service B".

1. Build and complete a dual input table: whether or not to use A in lines, use
   or not of B in columns. Add totals.
2. Calculate $P_A(B)$ and $P_B(A)$ For each calculation, specify the reference population used in
   denominator.
3. Interpret both probabilities with two separate sentences.

Fiscal year 5.2 · Production

A company manufactures objects in two workshops A and B, which provide respectively $\frac{1}{4}$ and $\frac{3}{4}$ of
Of the items produced by A, 4 percent are defective; among those produced by A, 4 percent are defective.
by B, 2 percent are. One randomly selects an object from the whole production.
D The event "the object is defective".

1. Represent the situation by a weighted tree and specify the meaning of each probability of the second
   level.
2. Calculate $P(D)$ and interpret the result as an overall default rate.
3. We now know that the chosen object is defective. What is the probability that it was manufactured?
   In Workshop A?

Exercise 5.3 · Independence to be verified

We are launching a balanced dice with six faces numbered from 1 to 6. A : "Provide
an even number" and B : "To obtain a number greater than or equal to 4".

1. Write out the outcome sets corresponding to A, B and $A \cap B$.
2. Determine whether events A and B are independent. Justify using a numerical criterion.
3. Explain why they are not incompatible either.

Fiscal year 5.4 · No remission

One urn contains three red balls and two blue balls, indistinguishable to the touch.
chance a first ball, then a second without putting back the first. At each draw, all the
balls present in the urn are equiprobable.

1. Build the weighted tree of the two prints, distinguishing the color obtained from the first and the second second draw.
2. Calculate the probability of getting two red balls.
3. Calculate the probability of getting one ball of each color. Explain why two paths of the tree must be taken into account.

Exercise 5.5 · Three tests

An experiment involves a success of probability $\frac{1}{3}$ and a failure. We repeat it three times in a way independent, under the same conditions.

1. Build a tree on three levels, or list eight possible outcomes.
2. Calculate the probability of achieving exactly a success. Indicate the paths chosen.
3. Calculate the probability of achieving at least two successes by distinguishing the cases of two and three success.

Exercise 5.6 · Detector

A device controls objects of which 2 percent are defective. It triggers a signal for 90
per cent of defective objects and, wrongly, 5 per cent of flawless objects.
It's a random, controlled object. D the event "the object is defective" and S the event « device triggers a signal".

1. Determine the proportion of objects actually defective among the objects reported. representation and detailed reasoning.
2. Give an exact value and then a percentage rounded to the tenth.
3. A person directly offers 90 percent as an answer. Explain precisely what confusion of probabilities it commits.

Exercise 5.7 · Together and Cardinal

We all work together. $E = \{1; 2; 3; 4\}$. Its parts are considered $A = \{1; 2\}$ and $B = \{2; 4\}$.

1. Write $A \cap B$, $A \cup B$ and the supplement to A in E. Give each one's cardinal.
2. Write all couples of $A \times B$then give his cardinal.
3. Explain why couples (1; 2) and (2; 1) cannot be confused. to $A \times B$?

Fiscal year 5.8 · Deferred review

In a random experiment, two events A and B check

$$P(A) = \frac{1}{2}, \qquad P_A(B) = \frac{1}{4}, \qquad P_{\overline{A}}(B) = \frac{3}{4}.$$

1. Calculate $P(B)$ using a tree or total probability formula.

2. Calculate $P_B(A)$ by showing the common event used in the numerator.

3. Probability $P_A(B)$ and $P_B(A)$ are they equal here ? Is this equality a general property ?

Personal work. Perform only the task assigned at the end of the meeting, in progress Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions Resuming exercise 8 a week after the chapter (10 min). is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, Exercise 67 p. 336, Questions 1, 2 and 3. Building a tree by keeping the D and N events in the manual. Attention: N is a negative test.

CQFD, exercise 70 p. 336, questions a and b. Count the four equiprobable outcomes of the two launches to refute the proposed equalitys.

MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $P_A(B)$ has as a denominator

   A. $P(B)$
   B. $P(A)$
   C. $P(A \cap B)$
   D. $1 - P(A)$

2. $P(A) = 0{,}4$, $P_A(B) = 0{,}2$: intersection

   A. $0{,}08$
   B. $0{,}6$
   C. $0{,}5$
   D. $0{,}2$

3. A partition includes events

   A. Equal
   B. All independent
   C. the same probability
   D. disjointed and universe-covered

4. Independence is equivalent to

   A. $P(A) = P(B)$
   B. $P(A \cap B) = 0$
   C. $P(A \cap B) = P(A)P(B)$
   D. $A = B$

5. Two incompatible events of positive probabilities are

   A. Independent
   B. Non-independent
   C. Equal
   D. necessarily complementary

MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. Three independent casts of a balanced piece, at least one stack:

   A. $\frac{1}{8}$
   B. $\frac{3}{8}$
   C. $\frac{7}{8}$
   D. $\frac{1}{2}$

2. Two independent tests with $p = \frac{1}{3}$, two successes:

A. $\frac{1}{9}$

B. $\frac{2}{3}$

C. $\frac{1}{3}$

D. $\frac{4}{9}$

3. The branches resulting from the same node have as their sum

A. 0

B. 2

C. Probability of the Path

D. 1

4. A simulation gives a frequency equal to 0.51.

A. Proofs $p = 0{,}51$

B. is fluctuating observation

C. Proofs $p = \frac{1}{2}$

D. Rebuttal $p = \frac{1}{2}$

5. At the draw without discount, after a red among 3 reds and 2 blues, probability of red:

A. $\frac{3}{5}$

B. $\frac{3}{4}$

C. $\frac{1}{2}$

D. $\frac{2}{5}$

True or false?

1. $P_A(B) = P_B(A)$ for all events of positive probabilities.

2. If A and B are independent, A and $\overline{B}$ So do I.

3. Two successive drawings are still independent.

Personal review

I find the right denominator; I check the amounts at the nodes; I justify independence; I count the paths without confusing them.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and a mistake to avoid.

Corrected and conduct of chapter 5

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. $\frac{12}{40} = \frac{3}{10} = 30\,\%$. *Action:* Partial and reference effective linkage.
2. $0{,}7$. *Action:* Represent the universe shared in two events.
3. $A \cap B$. *Action:* Coloring the intersection of two sets.
4. $A \cup B$. *Action:* Include common issues in the meeting.
5. $0{,}1$. *Action:* Replace the decimals with fractions.
6. $\frac{3}{6} = \frac{1}{2}$. *Action:* Identify the equiprobability hypothesis.

Research activity

Intersection required 12; boxes 12,28,18,42. Replace by 13,27,17,43 keeps margins but $\frac{13}{100} \neq 0{,}4 \times 0{,}3$.

Immediate applications

1. $\frac{5}{60} = \frac{1}{12}$ and $\frac{5}{20} = \frac{1}{4}$.
2. $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$.
3. No: $0{,}4 \times 0{,}5 = 0{,}20 \neq 0{,}18$.
4. $(\frac{2}{3})^4 = \frac{16}{81}$ then $1 - \frac{16}{81} = \frac{65}{81}$.
5. $\frac{675}{2000} = 0{,}3375$; this is an estimate, not the exact value.

Exercises: approach and results

Exercise 5.1 · A table. Boxes: 15 (A and B), 25 (A without B), 20 (B without A), 40 (Neither A nor B). $P_A(B) = \frac{15}{40} = \frac{3}{8}$; $P_B(A) = \frac{15}{35} = \frac{3}{7}$.

Exercise 5.2 · Production. $P(D) = 0{,}25 \times 0{,}04 + 0{,}75 \times 0{,}02 = 0{,}025$. $P_D(A) = \frac{0{,}01}{0{,}025} = 0{,}4$.
The denominator changes when it is conditioned by D.

Exercise 5.3 · Independence to be checked. $P(A) = P(B) = \frac{1}{2}$; intersection $\{4; 6\}$ probability $\frac{1}{-} \neq \frac{1}{-}$: The fact that another threshold choice gave independence does not transfer
Not automatically.

Exercise 5.4 · No remission. Two reds: ($\frac{3}{5})(\frac{2}{4}) = \frac{3}{10}$. One of each: ($\frac{3}{5})(\frac{2}{4}) + (\frac{2}{5})(\frac{3}{4}) = \frac{3}{5}$. The order distinguishes two disjointed paths.

Exercise 5.5 · Three tests. One success: three paths, each ($\frac{1}{3})(\frac{2}{3})^2 = \frac{4}{27}$Total $\frac{4}{9}$. Two success: $3(\frac{1}{3})^2(\frac{2}{3}) = \frac{2}{9}$; three : $\frac{1}{27}$; Total $\frac{7}{27}$.

Exercise 5.6 · Detector. $P(S) = 0{,}02(0{,}9)+0{,}98(0{,}05) = 0{,}067$. $P_S(D) = \frac{0{,}018}{0{,}067} = \frac{18}{67} \approx 26{,}9\,\%$. The initial low rate explains the difference with 90%.

Exercise 5.7 · Together and Cardinal. $A \cap B = \{2\}$ (1) ; $A \cup B = \{1;2;4\}$ (3) ; $E \setminus A = \{3;4\}$ (2); product $\{(1;2),(1;4),(2;2),(2;4)\}$ (4) A couple is ordered.

Exercise 5.8 · Deferred review. $P(B) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$. $P_B(A) = \frac{\frac{1}{8}}{\frac{1}{2}} = \frac{1}{4}$. Digital equality with $P_A(B)$ is particular to these data.

CQFD, Exercise 67 p. 336, Questions 1, 2 and 3. $P(\overline{D} \cap \overline{N}) = 0{,}9 \times 0{,}05 = 0{,}045$. Probability a positive test is 0 ,$1 \times 0{,}97 + 0{,}045 = 0{,}142$. Among the positive tests, the probability of not be doped is worth $\frac{0{,}045}{0{,}142} = \frac{45}{142} \approx 31{,}7\%$ Do not reverse positive and negative test.

CQFD, exercise 70 p. 336, questions a and b. $P(B) = \frac{1}{4}$, $P_A(B) = \frac{1}{2}$ and $P_{\overline{A}}(B)$ = 0. $\frac{1}{4} \neq \frac{1}{2}$ + 0, then 0 $\neq 1 - \frac{1}{2}$. The formula of total probabilities includes weights $P(A)$ and $P(\overline{A})$; the supplement applies to B when the same condition is laid down.

MCQ: answers and reasons

MCQ 1.

1. B. Reference universe A.
2. A. Product of branches.
3. D. Definition.
4. C. Criterion without division.
5. B. Zero intersection, positive product.

MCQ 2.

1. C. Supplement of three sides.
2. A. Output p^2.
3. D. Probabilities of conditional partition.
4. B. Observed frequency and probability of the model differ.
5. C. There are 2 reds on 4 balls.

True or false

1. False: 12 common objects out of 40 A and 30 B give $\frac{3}{10}$ and $\frac{2}{5}$

2. True: $P(A \cap \overline{B}) = P(A) - P(A \cap B) = P(A)(1 - P(B))$.

3. False: without remission, the composition of the urn changes.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.